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What's Backpropagation? - Mathematical Intuition



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Backpropagation using a Peanut Butter Sandwich

Imagine you're making a peanut butter sandwich for your friend.

Step 1: Forward Pass

You make the sandwich by:

- Taking two slices of bread
- Spreading peanut butter
- Closing the sandwich

Your friend tastes it and says:

It's too dry."

This feedback is like the **error** or **loss** in a neural network.

Step 2: Backpropagation

Instead of randomly making another sandwich, you think backwards.

You ask:

- Did I use too little peanut butter?
- Was the bread too thick?
- Did I spread it unevenly?

You trace the mistake back through each step that created the sandwich.



Step 3: Update

Based on the feedback, you make small changes:

- Add a little more peanut butter.
- Spread it evenly.
- Maybe use softer bread.

Then you make another sandwich.

You repeat this process until your friend says:

“Perfect!”

Connecting to Neural Networks

- Making the sandwich = **Forward pass**
- Friend's feedback = **Loss (error)**
- Thinking backwards to find what caused the error = **Backpropagation**
- Adjusting the recipe = **Updating the weights**
- Making another sandwich = **Next training iteration**

classmate

Date _____

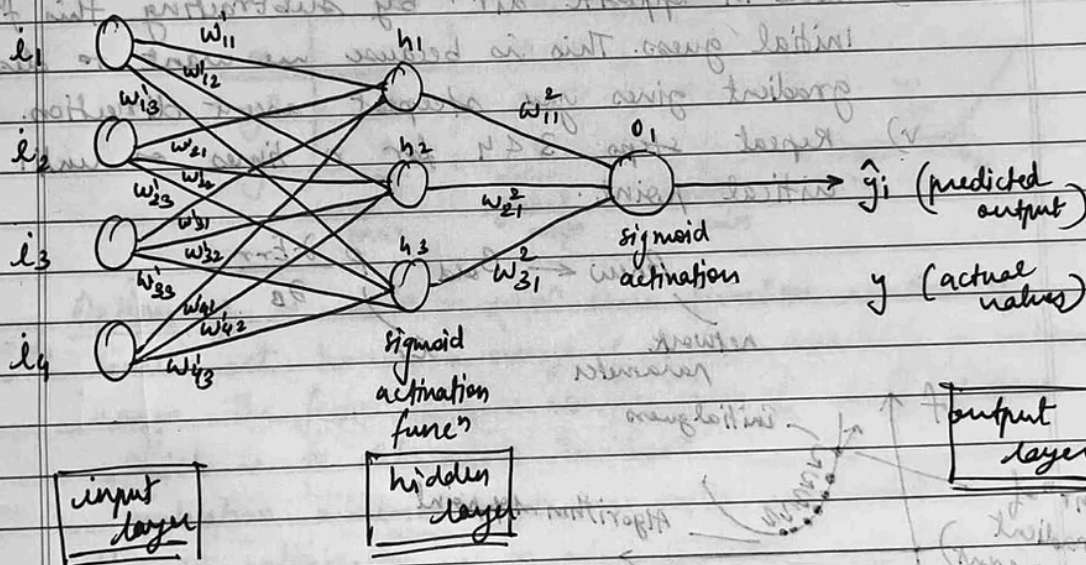
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Backpropagation

→ an algorithm that trains the weights of connecting nodes on layered neural network.

Forward phase: in which activations are propagated from the input layer to the output layer.

Backward phase: in which error between the observed and requested values in output layers is propagated backwards to adjust the weights and bias values.



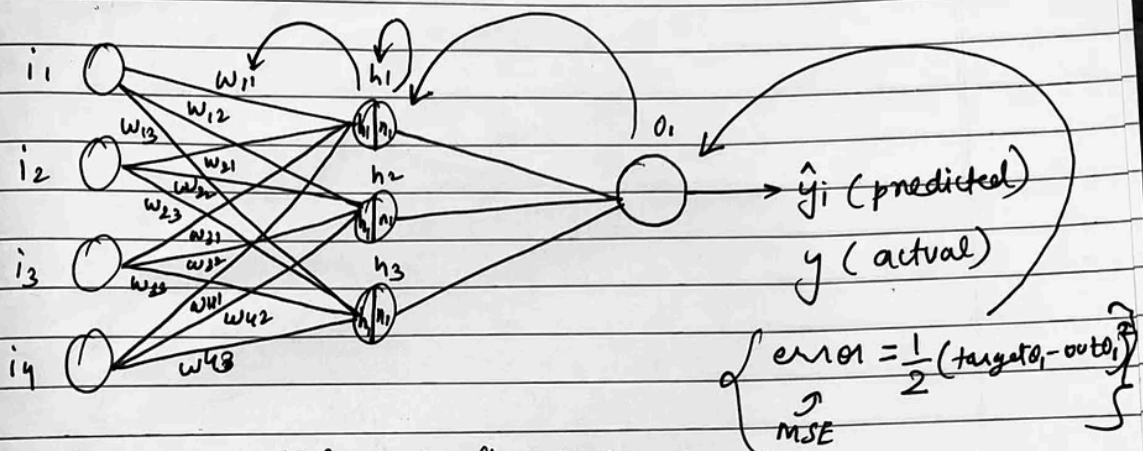
Architecture of nn.

For backpropagation we are gonna focus on the updation of weights after calculating the loss from (predicted outputs & actual values) $\hat{y}_i$ & y .

In our calculation we will only be updating weights.

Cost function: Basically we can control this function by changing weights & bias. The level of adjustment is determined by the gradients of the cost funcⁿ wrt parameters.

In backprop. algorithm, errors are propagated back from the output layer to the preceding layers. The backpropⁿ algorithm is applied for multiple-layer feedforward network.



we use partial derivative backwards -

$$\frac{\partial \text{error}}{\partial w_{11}} = \frac{\partial \text{error}}{\partial \text{out}_{o_1}} \times \frac{\partial \text{error}}{\partial \text{net}_{o_1}} \times \frac{\partial \text{error}}{\partial w_{11}} \quad [\text{chain rule}]$$

$$\frac{\partial \text{error}}{\partial w_{11}} = -(\text{target}_{o_1} - \text{out}_{o_1}) \times \text{out}_{o_1} (1 - \text{out}_{o_1}) \times w_{11}$$

in similar way we can find of $w_{21}, w_{31}, \dots, w_n$.
Then we update the weights:

$$\begin{aligned} w_{11}^2 &= w_{11}^1 - \alpha \frac{\partial \text{error}}{\partial w_{11}^1} \\ w_{21}^1 &= w_{21}^1 - \alpha \frac{\partial \text{error}}{\partial w_{21}^1} \end{aligned}$$



Written by Shetye Swara

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